

Analog Control of Inverted Pendulum

Demo and Full Report

Group 6

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March 29, 2026

1 Introduction and Motivation

The inverted pendulum is a useful control-system benchmark because the upright equilibrium is naturally unstable. Without feedback, even a small angular displacement from vertical grows rather than correcting itself. Stabilizing the pendulum therefore requires a controller that senses the pendulum position, estimates or measures its velocity, and commands a motor current in the direction that restores the pendulum toward the upright position.

The motivation for this project was to connect the theoretical design of a full-state feedback controller to an actual analog implementation. Instead of implementing the controller digitally, the feedback law was realized using sensor-conditioning circuitry, a practical differentiator, and an inverting summing amplifier. This made the project a direct test of how mathematical control gains translate into resistor ratios, voltage scalings, and hardware tuning.

The objective of the experiment was to design, simulate, build, and test an analog controller for the inverted pendulum. The report is organized as follows. First, the final schematic is presented to show the overall circuit architecture. Next, the position circuit, differentiator, and feedback-gain calculations are derived. The circuit implementation and closed-loop results are then shown, followed by a discussion of why the experimental response differed from the nominal theoretical target. The corrected demo video link is provided in the final section.

2 Final Schematic of Design

Figure 1 shows the final schematic of the analog control system. The design contains three main parts: the position-sensing circuit, the velocity-estimation circuit, and the control summing circuit. Presenting the full schematic first gives a high-level view of how the measured signals are conditioned and combined before being sent to the servo amplifier.

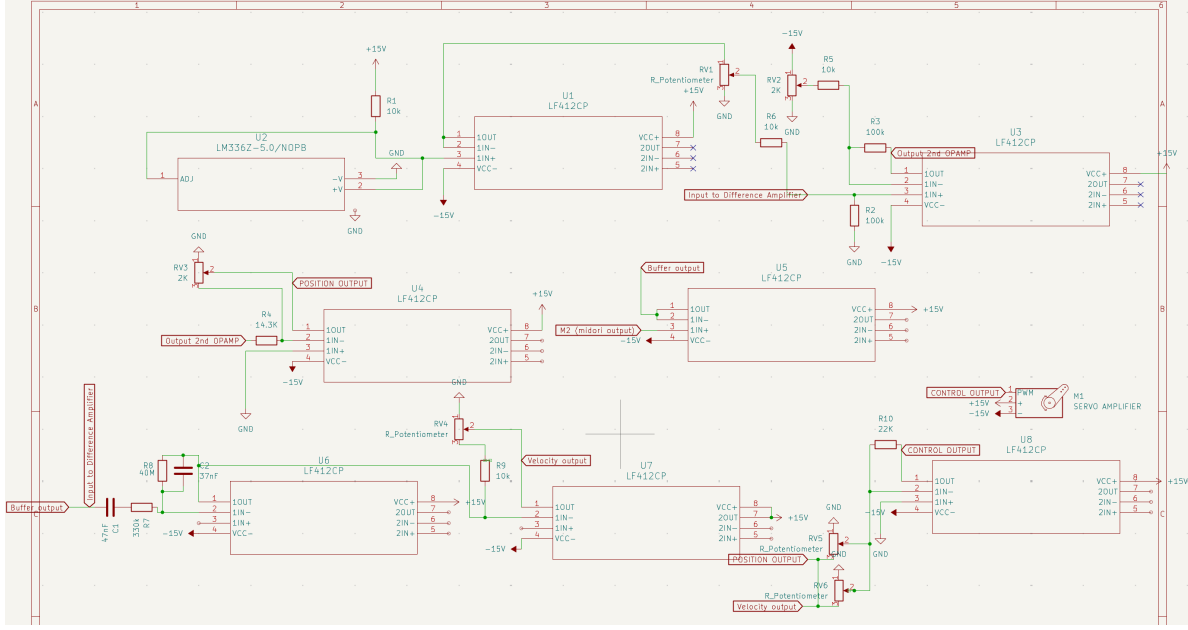


Figure 1: Final schematic of design including position, velocity, and control circuits.

3 Calculations

This section explains the calculations used to connect the desired control behavior to the analog circuit. The design process begins with the position measurement, continues with the practical differentiator used for velocity feedback, and ends with the full-state feedback gains and resistor ratios.

3.1 Position Circuit

We decided to run a preliminary simulation using TINA-TI, as this software has a vast library of Texas Instruments components. Unfortunately, there was no symbol available for the LM336 voltage regulator. Instead, we used a Zener diode with a breakdown voltage of 5 V, breakdown current of 1 mA, and a Zener resistance of 0.6Ω , which behaves similarly to the LM336 in practice.

The Midori potentiometer is represented as a standard potentiometer with power rating

$$P = 2\text{ W}, \quad V_{\text{max}} = 5\text{ V},$$

and set to 50%. Based on this configuration, we calculated that the input voltage e_0 at 0° is approximately 2.5 V, which corresponds to 50% of 5 V.

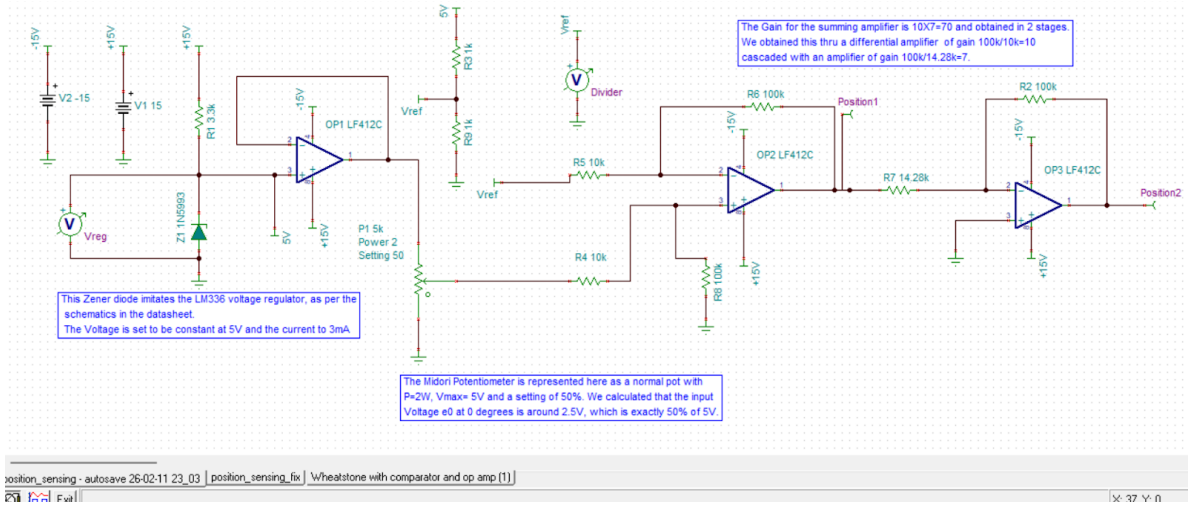


Figure 2: Preliminary simulation circuit using TINA-TI.

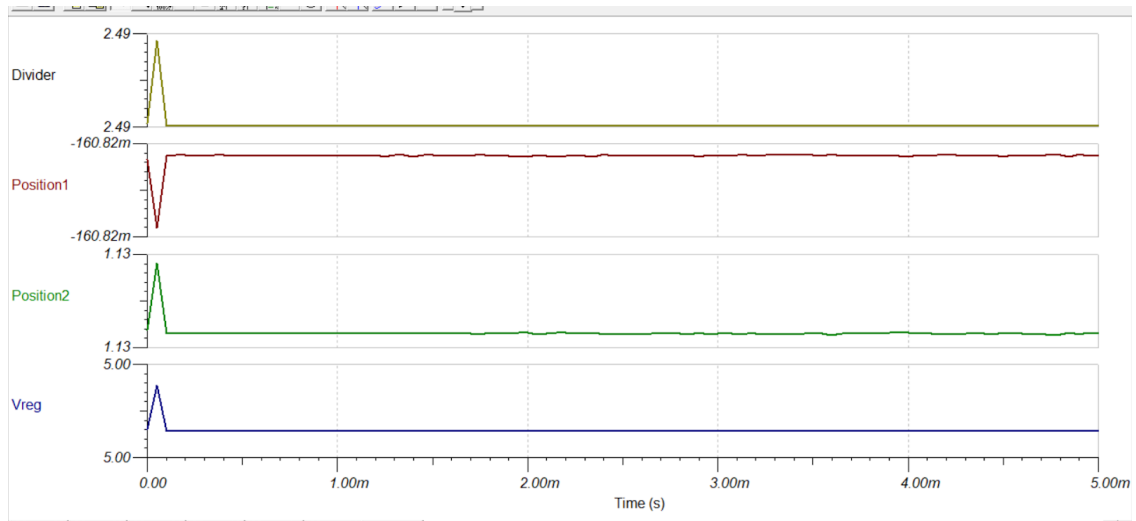


Figure 3: Plot of simulation at the nominal 50% potentiometer position.

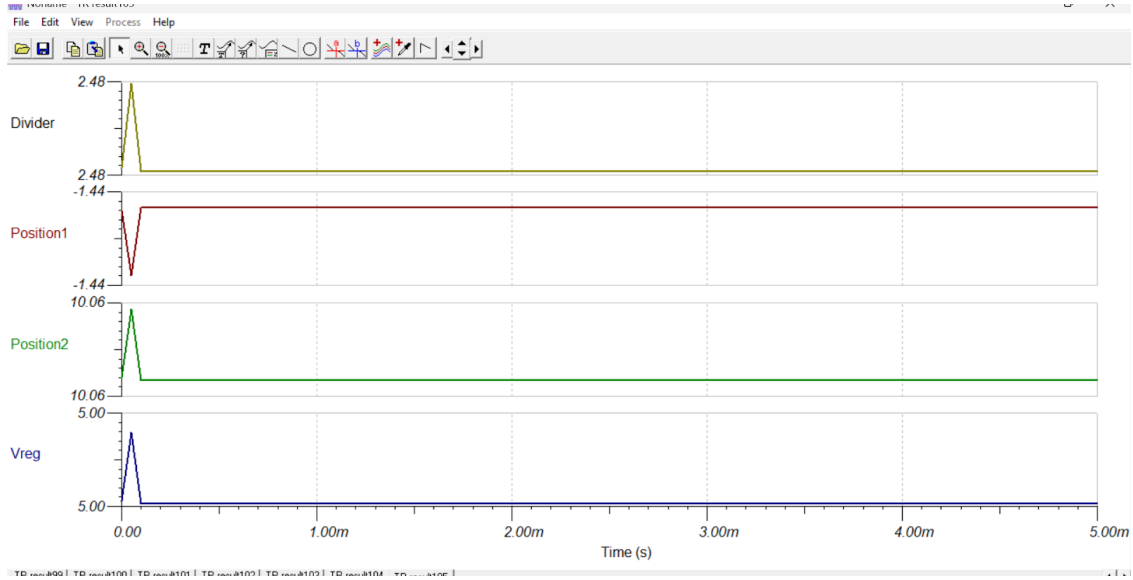


Figure 4: Plot of simulation at 47.3%.

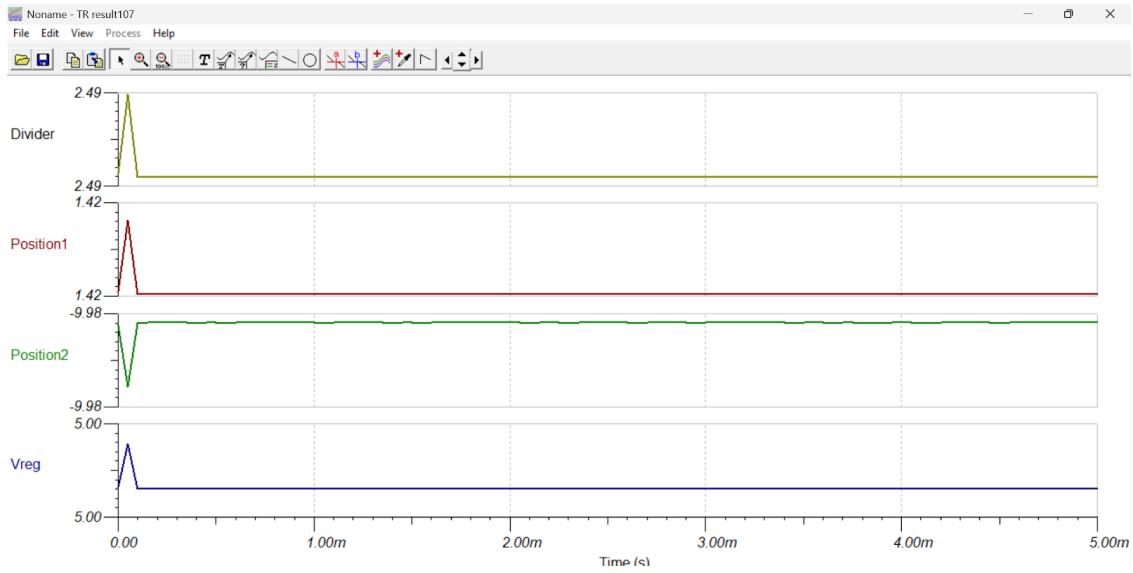


Figure 5: Plot of simulation at 53.35%.

3.2 Practical Differentiator

The velocity signal was produced using a practical differentiator rather than an ideal differentiator. This is important because an ideal differentiator amplifies high-frequency noise strongly, which would make the measured velocity signal noisy and difficult to use in feedback. The practical differentiator limits the useful frequency band so that it differentiates over the range of interest while rolling off the gain at higher frequencies.

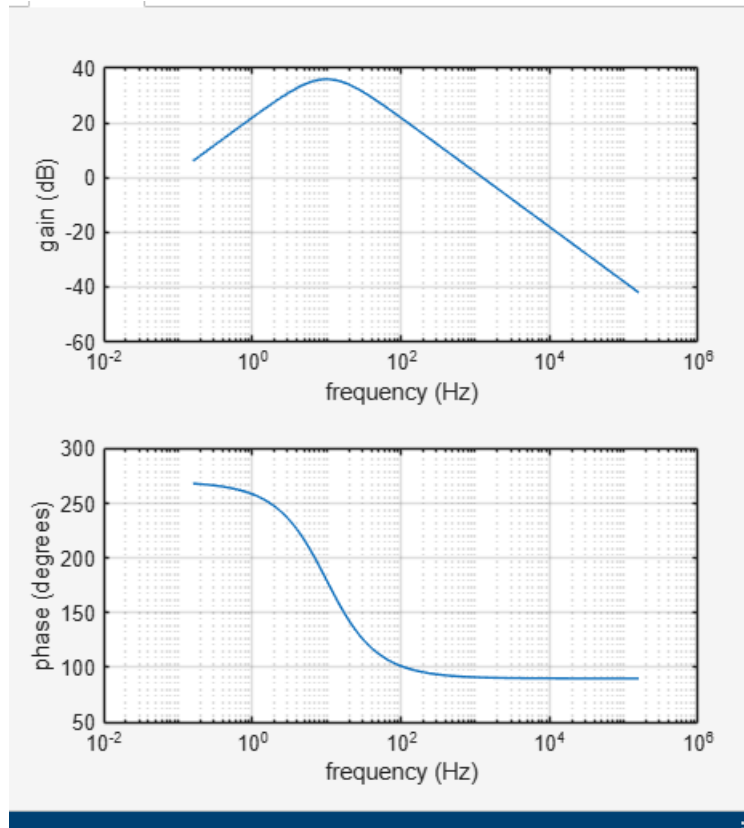


Figure 6: Gain and phase response of the practical differentiator circuit.

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What is the desired cutoff frequency in Hz? 10
What is the desired gain at 1 rad/sec? 2
Enter title of plot:
What is the desired value of the input capacitor in Farads? 47e-9

R1 =

    3.3863e+05|

R2 =

    4.2564e+07

C1 =

    4.7000e-08

C2 =

    3.7392e-10

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Figure 7: Calculation for resistor values for the practical differentiator.

3.3 Control Design for the Inverted Pendulum

We modeled the pendulum about the upright equilibrium using the linearized equation

$$m_1 l^2 \ddot{\theta} - m_2 g l \theta = K_\tau i, \quad (1)$$

where i is the motor current command and θ is the angular displacement from vertical. Using the measured masses

$$m_r = 0.0349 \text{ kg}, \quad m_b = 0.0462 \text{ kg},$$

and the geometry

$$l = 0.136 \text{ m}, \quad r = 0.0125 \text{ m}, \quad h = 0.012 \text{ m},$$

we obtain

$$m_1 = \frac{1}{3}m_r + m_b + \frac{1}{4}m_b \frac{r^2}{l^2} + \frac{1}{12}m_b \frac{h^2}{l^2} \approx 0.05796 \text{ kg}, \quad (2)$$

$$m_2 = \frac{1}{2}m_r + m_b \approx 0.06365 \text{ kg}. \quad (3)$$

Letting the state vector be

$$x = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix},$$

the state-space model becomes

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ a & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ b \end{bmatrix} i, \quad (4)$$

with

$$a = \frac{m_2 g}{m_1 l} \approx 79.21, \quad b = \frac{K_\tau}{m_1 l^2}. \quad (5)$$

Using $K_\tau = 0.0293 \text{ N} \cdot \text{m/A}$, we get

$$b \approx 27.33. \quad (6)$$

We chose full-state feedback of the form

$$i = -K_\theta \theta - K_\omega \dot{\theta}. \quad (7)$$

The closed-loop characteristic equation is then

$$s^2 + bK_\omega s + (bK_\theta - a) = 0. \quad (8)$$

To meet the design specifications, we used

$$\zeta = 0.707, \quad T_{s,2\%} = 0.5 \text{ s}.$$

Using the standard second-order settling time relation

$$T_{s,2\%} = \frac{4}{\zeta \omega_n}, \quad (9)$$

we obtain

$$\omega_n = \frac{4}{\zeta T_s} = \frac{4}{(0.707)(0.5)} \approx 11.32 \text{ rad/s}. \quad (10)$$

Thus the desired characteristic polynomial is

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = s^2 + 16.0s + 128.1. \quad (11)$$

Matching coefficients gives

$$bK_\omega = 2\zeta\omega_n = 16.0, \quad (12)$$

$$bK_\theta - a = \omega_n^2 = 128.1. \quad (13)$$

Therefore,

$$K_\omega = \frac{16.0}{27.33} \approx 0.585 \text{ A/(rad/s)}, \quad (14)$$

$$K_\theta = \frac{128.1 + 79.21}{27.33} \approx 7.58 \text{ A/rad}. \quad (15)$$

To implement this controller with analog electronics, we used the measured position and velocity voltages

$$e_\theta = H_\theta \theta, \quad e_\omega = H_\omega \dot{\theta}, \quad (16)$$

and an inverting summing amplifier

$$e_{\text{control}} = - \left(\frac{R_f}{R_{i\theta}} e_\theta + \frac{R_f}{R_{i\omega}} e_\omega \right). \quad (17)$$

The servo amplifier converts control voltage to current according to

$$i = K_s e_{\text{control}}, \quad K_s = 0.6 \text{ A/V}. \quad (18)$$

Using the design sensor scalings

$$H_\theta \approx 57.3 \text{ V/rad}, \quad H_\omega \approx 9.12 \text{ V/(rad/s)},$$

the required resistor ratios are

$$\frac{R_f}{R_{i\theta}} = \frac{K_\theta}{K_s H_\theta} = \frac{7.58}{(0.6)(57.3)} \approx 0.220, \quad (19)$$

$$\frac{R_f}{R_{i\omega}} = \frac{K_\omega}{K_s H_\omega} = \frac{0.585}{(0.6)(9.12)} \approx 0.107. \quad (20)$$

Choosing

$$R_f = 22 \text{ k}\Omega,$$

gives

$$R_{i\theta} \approx \frac{22 \text{ k}\Omega}{0.220} \approx 100 \text{ k}\Omega, \quad R_{i\omega} \approx \frac{22 \text{ k}\Omega}{0.107} \approx 206 \text{ k}\Omega. \quad (21)$$

In practice, we used nearby standard values and trim potentiometers to fine-tune the response experimentally.

4 Circuit Implementation

After deriving the component values, the controller was implemented on a breadboard and connected to the pendulum hardware. Figure 8 shows the physical circuit and the measurement setup. The labeled outputs made it possible to observe the position, velocity, and control signals during testing.

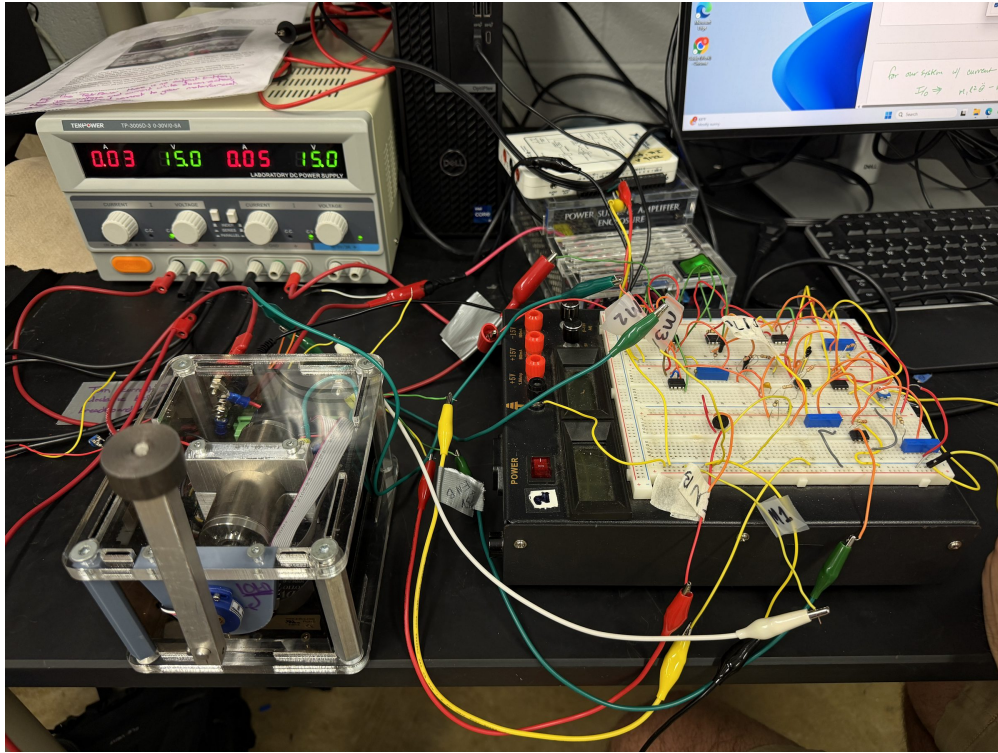


Figure 8: Circuit implementation with labeled outputs and connection wires. Some labels were difficult to angle for visibility, but most are clearly shown.

5 Results

The closed-loop control test is shown in Figure 9. The controller produced a restoring action when the pendulum was displaced from the upright position. However, the measured response still contained noticeable oscillation, indicating that the real system was more underdamped than the nominal second-order design target.

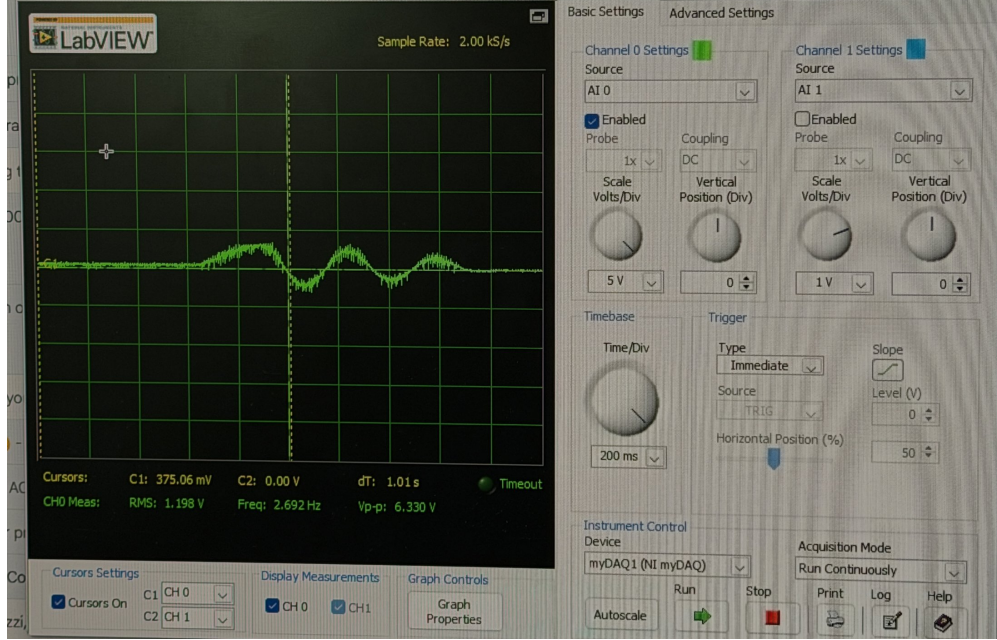


Figure 9: Closed-loop response of the control circuit with approximately ± 10 V control limits.

6 Discussion

6.1 Discussion of Closed-Loop Results

The closed-loop controller was able to generate a restoring action for small perturbations of the pendulum from the upright position, but the measured response did not fully satisfy the desired nominal performance of

$$\zeta = 0.707, \quad T_{s,2\%} = 0.5 \text{ s.}$$

Instead of showing a clearly decaying transient that returned rapidly to the upright equilibrium, the system remained noticeably underdamped. In particular, after a small displacement from vertical, the motor current and control signal continued to oscillate rather than settling cleanly to zero. This indicates that the implemented damping in the real system was weaker than predicted by the ideal model.

There are several likely reasons for this discrepancy between the theoretical design and the experimental behavior. First, the controller gains were implemented using trim potentiometers, so the realized electrical gains were only approximately equal to the calculated values. Any error in the ratios $R_f/R_{i\theta}$ and $R_f/R_{i\omega}$ directly changes the realized values of K_θ and K_ω . Second, the position and especially the velocity sensing circuits were not perfectly calibrated. The position channel was close to the desired scaling, but the velocity channel was more difficult to verify experimentally, and any error in the effective value of H_ω has a strong effect on the damping term. Since the velocity feedback is the primary source of damping in the closed-loop response, an inaccurate or underscaled velocity signal would make the system remain underdamped even if the position loop were approximately correct.

A third source of error is the difference between the idealized model and the physical hardware. The control law was derived from a linearized model valid only for small motion near the upright position. In practice, the motor, servo amplifier, potentiometer, and analog circuitry introduce offsets, noise, saturation, and component tolerances that are not included in the ideal equations. In

addition, the effective pendulum parameters depend on the measured shaft and bob masses and dimensions, which were only approximately known during the initial design process. Small errors in m_1 , m_2 , and the torque constant K_τ propagate into the state-space coefficients a and b , and therefore into the calculated feedback gains.

To better understand the closed-loop behavior, we adjusted K_θ and K_ω experimentally by changing the trim potentiometers associated with the control summing amplifier. Increasing K_ω increased the influence of the velocity feedback term and generally made the response more damped. Decreasing K_ω reduced the damping and caused the pendulum to oscillate more strongly, producing a more underdamped response. In contrast, changing K_θ mainly affected how strongly the controller reacted to angular displacement. If K_θ was too small, the restoring action was weak and the pendulum returned slowly. If K_θ was made too large without a corresponding increase in K_ω , the system became more oscillatory. Thus, the experiments confirmed the expected roles of the two gains: K_θ primarily sets the restoring strength, while K_ω primarily sets the damping.

Overall, the experimental results were qualitatively consistent with the control design, since the pendulum responded to position and velocity feedback in the expected direction. However, the nominal target response was not fully achieved. The main limitation was that the real system remained more underdamped than desired. The most likely causes were imperfect calibration of the velocity sensing channel, uncertainty in the realized potentiometer-based control gains, and nonideal effects in the servo amplifier and analog electronics. Further improvement would require a more careful calibration of H_ω , direct measurement of the realized gain ratios in the control circuit, and finer tuning of K_ω to achieve the intended damping ratio.

7 Video Link

The demonstration video of the analog inverted-pendulum controller is available at the corrected link below:

[Analog Control of Inverted Pendulum Demo](https://youtu.be/v_uw70nvvIg?si=VsroG_AgkkgZ3enn)

https://youtu.be/v_uw70nvvIg?si=VsroG_AgkkgZ3enn